MIL-S-7780B(AER)
30 MARCH 1956
Superseding
MIL-S-7780A
1 December 1952

MILITARY SPECIFICATION

STARTERS; ELECTRIC, CONSTANT CURRENT, DC

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope.- This specification covers electric variable-voltage direct-current starters for operation from a constant-current power source for starting turbojet and turbo-prop engines.

1.2 Classification.- Starters shall be of the following types:

Type E-3: Starters having a rated output drive speed of 850 rpm and a rated output torque of 135 pounds-feet.

Type E-4: Starters having a rated output drive speed of 2000 rpm and a rated output torque of 61 pounds-feet.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| NN-P-515 | Plywood, Container Grade |
| NN-B-631C | Boxes, Wood, Wirebound, (For Domestic Shipment) |

Military

| | |
|---|---|
| MIL-D-5028 | Drawings and Data Lists; Preparation of (for Engines, Accessories and Other Auxiliary Equipment) |
| MIL-D-5480 | Data; Engineering and Technical (Reproduction Thereof) |
| MIL-L-6880 | Lubrication of Aircraft, General Specification for |
| MIL-P-116 | Preservation, methods of |
| MIL-P-6906 | Plates, Information and Identification |
| MIL-S-7742 | Screw Threads, Standard, Aeronautical |
| MIL-B-18364(AER) | Bulletins; Aircraft Power Plant, Propeller, Auxiliary Power and Starting Unit; Preparation of |
| JAN-A-669 | Anti-Seize Compound, White Lead Base, General Purpose (for Threaded Fittings) |
| JAN-P-106 | Packaging and packing, for overseas shipment, boxes, wood nailed |

STANDARDS

| | |
|---|---|
| MIL-STD-129 | Marking of Shipments |



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DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

| | |
|---|---|
| AND10398 | Metals - Definition of Dissimilar |
| AND20002 | Drive - Type XII Engine Accessory |
| AND20004 | Drive - Type XIV Engine Accessory |

PUBLICATIONS

Air Force-Navy Aeronautical Bulletins

| | |
|---|---|
| No. 143 | Specifications and Standards; Use of |
| No. 166 | Colors; List of Standard Aircraft Glossy |
| No. 182 | Material Changes and Substitutions; Aircraft Engine Parts (Production Contracts) |
| No. 343 | Specifications and Standards Applicable to Aircraft Engines and Propellers; Use of |
| No. 391 | Changes; Engineering, to Aircraft Engines, Propellers, and Aeronautical Equipment in Production and Service |

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT).

3. REQUIREMENTS

3.1 Qualification.- The starters furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Materials.- Materials used in the manufacture of starters shall be of high quality, suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are released by the Government and contain provisions for adequate tests. The use of contractor's specification will not constitute waiver of Government inspection.

3.2.1 Metals.- All metals used in the construction of starters, except metals which are in constant contact with lubricating oil, shall be of a corrosion-resisting type or shall be suitably protected to resist corrosion. The use of dissimilar metals, as defined by Drawing AND10398, shall be avoided wherever practicable.

3.2.2 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.2.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction.- The starters shall be designed to operate from a variable-voltage constant-current power source. They shall be of compact design and of such construction as to withstand vibration and other severe usage encountered under service conditions. The starters shall be capable of being operated throughout a temperature range of -55° to +70°C (-67° to +158°F), and shall not be damaged when exposed to temperatures from -75° to +135°C (-103° to +275°F).



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3.3.1 Environmental conditions.- Consideration shall be given in the design of starters for satisfactory operation during and after exposure to any combination of the following conditions in world-wide operation: Humidity, fungus, sunshine, rain, snow, sleet, hail, ice-fog, fog, rilluew, salt spray, ice, ozone, smoke, wind, sand, and dust.

3.3.2 Radio interference.- The starter shall be designed to minimize the generation of radio interference. Enclosing case construction shall provide continuity of electrical shielding with a low radio-frequency impedance path to ground and across all mechanical discontinuities.

3.3.3 Mounting and drive provisions.-

3.3.3.1 Quick detachable mounting.- Starters shall be provided with a quick detachable mounting of a design approved by the procuring activity. This mounting shall not require special tools for its use. It shall be automatically locked when on the engine, and there shall not be any unattached parts, such as clamp rings, bolts, or nuts when the starter is removed from the engine. Any adapter required for modifying the engine accessory drive in order to mount the starter shall be furnished with the starter, but it shall be considered as permanently attached to the engine after initial installation. The design of the quick-detachable mounting shall be such as to permit installation and removal of the starter when there is access to the starter at any one position.

3.3.3.2 Accessory drive.- Type E-3 starters shall be suitable for mounting on Drawing AND20002 type XII-F accessory drive provided with a 3-tooth jaw adapter. Type E-4 starters shall be suitable for mounting on Drawing AND20004 type XIV-A accessory drive.

3.3.3.3 Engagement.- Satisfactory engagement of the starter with the engine drive shall be independent of the voltage and current supplied to the starter and of the rate of voltage and current rise, provided the voltage and current are sufficient to cause rotation of the armature.

3.3.3.4 Disengagement and reengagement.- Disengagement shall be rapid, positive and complete at the end of the starting cycle. There shall be a minimum of ratcheting of the jaw caused by armature coasting. Reengagement of the starter after interruption during a starting cycle shall be accomplished without damage.

3.3.4 Rotation.- Drive rotation for starters shall be right-hand unless otherwise specified by the procuring activity. The rotation referred to is that of the starter jaw viewed from the anti-jaw end of the starter.

3.4 Components.- The starter shall consist of a variable voltage d-c motor, a gear reduction unit, and an engaging and automatic disengaging mechanism for meshing with the engine drive. A torque-limiting device may be included if considered desirable or necessary.

3.4.1 Motor.- The starter shall be provided with an ungrounded motor. Electrical connections to the starters shall be made to two insulated terminals. The motor shall be designed to operate from a variable-voltage constant-current power source which provides an initial voltage not greater than 8 volts at the motor terminals and an initial current of 1000 plus 100, minus zero, amperes. As the starter motor accelerates in speed the current is maintained at 1000 amperes until the voltage at the motor terminals is 30 volts, plus zero, minus 1 volt, at which time the current begins to reduce to 200-250 amperes where it is cut off by an undercurrent relay. During the time the current is reducing from 1000 amperes, the terminal voltage at the motor increases until it is 33-3/4 to 34 volts at the time the starter is cut off.

3.4.2 Starter oil seal.- The starter shall be designed to prevent seepage of oil from the engine into the starter and leakage of any appreciable quantity of lubricant from the starter.

3.4.3 Size.- The size of each starter shall not exceed the over-all dimensions specified on figure 1.



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3.4.4 Weight.- The total weight of the starters including quick detachable mounting, adapter, etc., shall not exceed the following:

Type E-3 Starter: 45.5 pounds
Type E-4 Starter: 45.5 pounds

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Changes in design and construction.-

3.6.1 Material substitutions and variations.- Temporary material substitutions and variations shall be made in accordance with ANA Bulletin No. 182.

3.6.2 Changes in design.- No changes shall be made in the design or materials of a starter model which has successfully completed the Qualification tests, except when such changes are approved by the following methods.

3.6.2.1 Class I changes.- Changes of this class are those changes affecting contract requirements covering weight, performance, durability, cost, or delivery dates of complete starters, or interchangeability of either parts or complete starters. Changes of this class shall be submitted in triplicate on an Engineering Change Proposal Form substantially as shown on figure 1 of ANA Bulletin No. 391, together with substantiating information and drawing clearly illustrating the proposed change and showing the effect on parts already in service and possible use of replaced parts, to the procuring activity for approval.

3.6.2.2 Class II changes.- Changes of this class are minor, do not affect those factors mentioned in paragraph 3.6.2.1, and may be approved by the procuring activity. These changes shall be forwarded in triplicate to the procuring activity for information after approval.

3.6.3 Service bulletin.- When requested by the procuring activity at the time of approval of a change, the contractor shall submit a service bulletin prepared in accordance with Specification MIL-S-18364(AER) in order to permit the procuring activity to incorporate the change in starters previously delivered.

3.6.4 Parts list.- The parts list of the starter which successfully completes the Qualification test shall constitute a requirement for production starters. Departure from the approved starter parts list in features of design and construction shall be governed by the provision herein for changes in design.

3.6.5 Responsibility for changes.- Approval by the procuring activity of changes from the Qualification test starter does not relieve the contractor of full responsibility for the results of such changes on any starter characteristic.

3.7 Protective treatment, coatings, and paint finishes.- Unless specifically excepted in ANA Bulletin No. 343, SAE Aeronautical Materials Specifications (AMS) for protective treatments, coatings, and paint finishes, may be used in lieu of Government specifications selected in accordance with ANA Bulletin No. 143 (see paragraph 3.2.2).

3.7.1 Protective treatment and coatings.-

3.7.1.1 Steel parts.- With the exception of the parts listed below, all exterior steel parts, and other steel parts subject to corrosion and not in constant contact with oil, shall be suitably protected by a process approved by the procuring activity.

(a) Corrosion-resisting steel parts
(b) Cable
(c) Tin-coated wire
(d) Members or portions of members which act as bearings or journals.



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3.7.1.2 Aluminum parts.- With the exception of the parts listed below, all exposed aluminum and aluminum-alloy parts shall be suitably protected by a process acceptable to the procuring activity. Other parts may also be excepted when application of treatment is considered impractical or unnecessary and when omission of treatment is specifically approved by the procuring activity.

(a) Metal sprayed surfaces

(b) Surfaces which are in constant contact with oil

(c) Unalloyed aluminum and aluminum-clad aluminum alloy

(d) Parts fabricated from 2s, 3s, 52s, and 61s, aluminum alloys (when painted).

3.7.1.3 Magnesium parts.- All magnesium-alloy parts shall be surface-treated for protection against corrosion.

3.7.2 Paint finishes.- With the exception of the parts listed below, all exposed metal surfaces shall be painted with one coat of primer and a minimum of two full wet finish coats, applied in such manner that no pinholing, holidays, sags, or runs, are encountered; and with the exception that magnesium-alloy parts shall receive two coats of primer and a minimum of two finish coats as described above. Additional parts may be excepted when application of paint finishes on any part thereof is considered impracticable or unnecessary, and when omission of any or all paint finishes is specifically approved by the procuring activity.

(a) Metal sprayed surfaces

(b) Corrosion-resisting steel, brass, copper or bronze parts

(c) Cable

(d) Working surfaces

(e) Threads

(f) Oil holes

(g) Cadmium-plated parts

(h) Unalloyed aluminum-clad aluminum alloy

(i) Parts fabricated from 2s, 3s, 52s, and 61s, aluminum alloy (when anodized)

(j) Surfaces of casting identification pads when suitably protected from corrosion by other means.

3.7.2.1 Primer coat.- The primer coat shall be a zinc-chromate primer. The primer shall be applied as soon as possible after prior surface treatments or coating. The primer coats shall be thoroughly dried prior to application of the finish coat(s) in order to prevent any consolidation of the primer or finish coats. When the primer coat is soiled or damaged by intervening operations between priming and finish coats, it shall be thoroughly cleaned and another light coat of primer added before the finish coat is applied.

3.7.2.2 Finish coat.- The finish coat for the basic starter shall be heat-resistant enamel or aluminum-pigmented varnish. Enamel other than that of aluminum color, shall match the applicable color of the Air Force-Navy aircraft color standards (glossy) and ANA Bulletin No. 166.



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3.9 Lubrication.- The component parts of the starter requiring lubrication shall be properly lubricated in accordance with Specification MIL-L-6880 when shipped and shall operate satisfactorily when in this condition.

3.9 Threaded parts.-

3.9.1 Screw threads.- Screw threads shall conform to Specification MIL-S-7742.

3.9.2 Coating threaded parts.- All threaded parts, except electrical terminals and safety-stop nuts, shall have a thin coating of compound conforming to Specification JAN-A-669 applied to the threads. No more of this compound shall be applied than is necessary to cover the surface of the threads.

3.9.3 Safetying.- Provisions shall be made to prevent all screws and screw parts from accidental loosening, by safety wiring or fitting with safety-stop nuts, where applicable. All other connections liable to be loosened by vibration shall be safetied in a similar manner.

3.9 Performance.- The starter shall satisfy the performance requirements specified in Section 4.

3.11 Identification of product.-

3.11.1 Nameplate.- Each starter shall be clearly and permanently marked with an irreversible nameplate conforming to Specification MIL-P-6906, securely attached to a part of the starter which will not ordinarily be renewed during normal service life, specifying the following information:

STARTER; ELECTRIC CONSTANT CURRENT, DC
Specification: MIL-S-7780B(AER)
Type
Stock No.
Jaw rotation (indicate by arrow)
Maximum voltage:_____volts
Manufacturer's Part No.
Contract or Order No.
Manufacturer's Serial No.
Manufacturer's name or trade-mark
U S Property

3.11.2 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.11.3 Marking of parts.- Each part and assembly shall be marked with its part number, which shall be the same as the drawing number, except the following.

3.11.3.1 Those parts which do not have a suitable or sufficient surface for a part number.

3.11.3.2 Those parts which are permanently assembled by welding, brazing, soldering, or riveting shall carry the assembly part number.

3.11.3.3 Part number.- The part number shall, when practicable, be located to permit its being read after assembly of the complete unit. The marking shall be such that it will not be effaced or obliterated as a result of service usage during the life of the part.

3.12 Workmanship.- All surfaces shall have a smooth finish and all details of workmanship shall be in accordance with high-grade aircraft starter manufacturing practice.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of starters shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on starters manufactured and submitted for acceptance under contract.

4.2 Qualification tests.-

4.2.1 Sampling instructions.- The Qualification test samples shall consist of two starters of each manufacturer's part number upon which qualification is desired. One set of assembly and detailed drawings shall be furnished with each new model starter submitted for the Qualification test. The assembly drawing shall show cutaway section showing all details in their normal assembled position and shall carry part numbers of all details and subassemblies. Samples shall be forwarded to the Supply Officer, Naval Air Material Center, U. S. Naval Base Station, Philadelphia 12, Pennsylvania, Attention: Superintendent, Aeronautical Engine Laboratory, and plainly identified by securely attached durable tags marked with the following information:

Sample for Qualification Tests
Starter: Electric, Constant Current DC
Manufacturer's Part Number
Name of Manufacturer
Project Number (Obtain from procuring activity)

Submitted by (name of manufacturer) (date) for Qualification test in accordance with Specification MIL-S-7780B(AER) under authorization (reference letter authorizing test)

4.2.2 Tests.- The Qualification tests of starters shall consist of all the tests of this specification as described under "Test methods". If considered desirable, the Qualification tests may be supplemented with tests under actual service conditions. The testing activity reserves the right to disassemble the starter partially or completely before, during, or after the Qualification tests.

4.3 Inspection tests.- The Inspection tests shall consist of Individual tests.

4.3.1 Individual tests.- Each starter shall be subjected to the following tests, as described under "Test methods" of this specification:

(a) Examination of product

(b) Overspeed

(c) Torque-speed output

In addition, the starters shall be subject to any other tests specified herein which the Inspector considers necessary to determine conformance with the requirements of this specification.

4.3.2 Rejection and retest.- Any starter failing to meet the requirements of the Individual tests shall be rejected and returned at the contractor's expense. Starters which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished to the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.



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4.4 Test conditions.- Unless otherwise specified herein, all tests shall be conducted at a temperature within the range of 10 to 35°C (50 to 95°F), and the electrical power source shall have the characteristics specified in paragraph 3.4.1 herein. The order in which the various tests are conducted shall be at the discretion of the test activity.

4.5 Test methods.-

4.5.1 Examination of product.- Each starter shall be carefully examined to determine conformance with this specification with respect to all the requirements not covered by specific tests.

4.5.2 Torque-speed Output.- The starter shall be tested to determine conformance with the Torque-Speed output requirements of table I with a current input of 1,000 amperes where applicable.

TABLE I

Torque-Speed Output

| Type | Volts +0, -1 | Jaw Speed RPM (Min) | Continuous Torque lb-ft | No-load Speed RPM (Min) | Max. Allowable Torque lb-ft |
|---|---|---|---|---|---|
| E-3 | 30 | 850 | 135±2 | 1600 | 450 |
| E-4 | 30 | 2000 | 61±1 | 3500 | 450 |

4.5.3 Maximum allowable torque.- The starter shall be tested to determine that the maximum torque output, including the impact at engagement, does not exceed the values specified in Table I with a current input of 1000 amperes. This test shall be conducted at the temperatures specified in paragraphs 4.4, 4.5.8.1 and 4.5.8.2.

4.5.4 Continuous torque.- The starter shall be checked to determine that the output speed is not less than that specified in Table I with a starter terminal voltage of 29 to 30 volts, an input current limited to a maximum of 1000 amperes, and an applied continuous torque within the range specified in Table I.

4.5.5 Efficiency.- The over-all efficiency of the starter at rated torque shall not be less than 68 percent at an input voltage of 30 volts, plus 0, minus 1.

TABLE II

Endurance Test

| Type | Flywheel Polar Moment of Inertia lb-ft$^2$ | PHASES A: RPM to which flywheel shall be accelerated within 8 sec. | B: 10 seconds at following conditions: RPM | B: Torque lb-ft (Min) | C: RPM to which flywheel shall be accelerated within 12 seconds |
|---|---|---|---|---|---|
| E-3 | 370 | 850 | 850 | 135 | 1340 |
| E-4 | 80 | 1880 | 1880 | 61.0 | 2950 |



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4.5.7 Overspeed.- The starter shall be subjected to a no-load test for a period of 60 seconds with 35.0 volts dc applied at the starter terminals. No failure or change in efficiency under rated load conditions shall occur as a result of this test.

4.5.8 Extreme temperature tests.-

4.5.8.1 Low temperature.- The starter shall be maintained at a temperature of -55°C (-67°F) for a minimum period of 8 hours and then operated for three consecutive endurance cycles. The output of the starter on each cycle shall be as specified in Table II with a current input of 1000 amperes where applicable.

4.5.8.2 High temperature.- The starter shall be maintained at a temperature of 70°C (158°F) for a minimum period of 8 hours and then operated for three consecutive endurance cycles. The output of the starter on each cycle shall be as specified in Table II with a current input of 1000 amperes where applicable.

4.5.9 Consecutive operation.- The starter shall be operated for five consecutive cycles without damage. Each cycle shall be under the Test Conditions specified in paragraph 4.4 and shall be equivalent to an endurance test cycle as defined in paragraph 4.5.6. Cooling air and rest periods shall not be permitted during this test.

4.6 Acceptance stamp.- All accepted starters shall be stamped with Inspector's official acceptance stamp.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.-

5.2.1 Level A.- Unless otherwise specified, each starter shall be packaged for shipment in accordance with specification MIL-P-116. Method IIb of this specification shall be employed for all starters designated as installation starters. Method IId of this specification shall be employed for all starters designated as spare starters.

5.2.2 Level C.- Starters shall be packaged in accordance with standard commercial practice.

5.3 Packing.- Shipping containers in-so-far as practicable, shall contain an identical number of articles, shall be of a uniform size, and shall be designed to enclose the contents in a snug, tight-fitting manner. When packed for shipment, the gross weight of the exterior container shall not exceed 200 pounds. Starters placed in metal cans or wood boxes do not require overpacking.

5.3.1 Level A.- Starters packaged in accordance with paragraph 5.2.1 shall be packed in an exterior shipping container in accordance with Federal specification PPP-B-601 (overseas type) or JAN-P-106. Plywood if used shall conform to type II class 1 or 2 of Federal specification NN-P-515.

5.3.2 Level B.- Starters packaged as specified in paragraph 5.2.1 shall be packed in exterior shipping containers conforming to Federal specifications PPP-B-601 (domestic type) or NN-B-631.

5.3.3 Level C.- The interior packages shall be placed in substantial commercial exterior shipping containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment. The use of corrugated or solid fiberboard having a minimum Mullen test of less than 275 pounds is prohibited.



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5.4 Marking.- Each package and exterior shipping container shall be marked to comply with Specification MIL-STD-129 and to include the following:

STARTER; ELECTRIC, CONSTANT CURRENT, DC
Specification MIL-S-7780B(AER)
Type
Date of Manufacture

6. NOTES

6.1 Intended use.- These starters are intended for use in starting aircraft turbojet and turbo-prop engines.

6.2 Ordering data.- Requisitions, contracts, and orders should state this specification number, the type designation of starter desired and level of packaging and packing required. Agreement should be made as to whether Government furnished containers will be furnished.

6.2.1 Drawings.- A starter contractor shall, as soon as practicable after the award of a contract for starters conforming to this specification, furnish the following drawings for each model starter on contract, unless the latest revisions have been previously submitted.

6.2.1.1 Drawing requirements.- The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted, must be in accordance with the requirements of Specifications MIL-D-5028 and MIL-D-5480.

6.2.1.1.1 Vandyke drawings.- One complete set of detail, assembly, subassembly, and installation vandyke drawings shall be sent to the following destination:

U. S. Naval Air Station
Johnsville, Pennsylvania
Attention: Technical Records Division

6.2.1.1.2 Blueprints.- One complete set of blueprints or equivalent made from the above-mentioned vandyke drawings shall be sent to each of the following destinations:

Bureau of Aeronautics
Department of the Navy
Washington 25, D. C.
Attention Plan Files

Aviation Supply Office
Philadelphia, Pennsylvania

6.2.1.1.3 Microfilm.- One complete set of detail, subassembly, assembly, and installation drawings on microfilm shall be sent to the following destination:

Bureau of Aeronautics
Department of the Navy
Washington 25, D. C.
Attention: Power Plant Division

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Supply Officer, Naval Air Material Center, the activity responsible for qualification, U. S. Naval Base Station, Philadelphia 12, Pennsylvania, Attention:



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Superintendent, Aeronautical Engine Laboratory, with copies to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Navy tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the above designated activity responsible for qualification.

6.3.1.1 It is to be understood that the starters supplied under contract shall be identical in every respect to the sample tested and found satisfactory and to the detail drawings thereof, except for changes previously approved as specified herein. An unapprpved change from the qualification sample or from the drawing shall constitute sufficient cause for rejection.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

*(TO DETACH THIS FO[ CUT ALONG THIS LINE.)*

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